FUNDAMENTALS OF SPEECH RECOGNITION

A detailed explanation

SPEECH RECOGNITION

Speech recognition is the technology that enables a machine or computer to identify and process human speech into a format that it can understand and act upon.

This involves converting spoken language into text or commands.

The process typically includes capturing audio signals, processing these signals to extract features, and then using algorithms to match these features to known patterns of speech.

BRIEF HISTORY AND EVOLUTION

The journey of speech recognition technology began in the mid-20th century:

- **1950s-1960s**: Early research focused on recognizing digits and small vocabularies. The first notable system was the **Audrey system** by Bell Labs in 1952, which could recognize digits spoken by a single voice.

- **1970s-1980s**: The development of **Hidden Markov Models (HMMs)** significantly improved the accuracy and robustness of speech recognition systems. IBM's **Tangora** system in the 1980s could recognize a vocabulary of about 20,000 words.

- **1990s-2000s**: The advent of **statistical models** and **machine learning** techniques further enhanced speech recognition capabilities. Systems like **Dragon NaturallySpeaking** became commercially available, allowing users to dictate text and control their computers by voice.

- **2010s-Present**: The rise of **deep learning** and **neural networks** has revolutionized speech recognition. Technologies like **Google Assistant**, **Amazon Alexa**, and **Apple Siri** leverage advanced neural network architectures to achieve near-human levels of accuracy.

HISTORY

https://medium.com/swlh/the-past-present-and-future-of-speech-recognition-technology-cf13c179aaf

IMPORTANCE AND RELEVANCE

Speech recognition technology has become increasingly important and relevant due to its wide range of applications:

Accessibility: It provides essential tools for individuals with disabilities, enabling them to interact with technology through voice commands.

Productivity: Speech-to-text applications help professionals transcribe meetings, create documents, and manage tasks more efficiently.

Consumer Devices: Voice-activated assistants like Alexa, Siri, and Google Assistant have become integral parts of smart homes, allowing users to control devices, get information, and perform tasks hands-free.

Customer Service: Automated customer service systems use speech recognition to handle inquiries, provide support, and improve customer experience.

Healthcare: Medical professionals use speech recognition for dictating patient notes, reducing the time spent on documentation and allowing more focus on patient care.

VIDEO

https://www.youtube.com/watch?v=uSNUmJffK4c

https://www.youtube.com/watch?v=PWVH3Vx3dCI

SAMPLING THE AUDIO

Sampling is the process of converting a continuous audio signal into a discrete digital signal by taking measurements at regular intervals.

The rate at which these measurements are taken is called the **sampling rate**.

Common sampling rates include 44.1 kHz (used in CDs) and 48 kHz (used in professional audio equipment). The higher the sampling rate, the more accurately the digital signal represents the original audio.

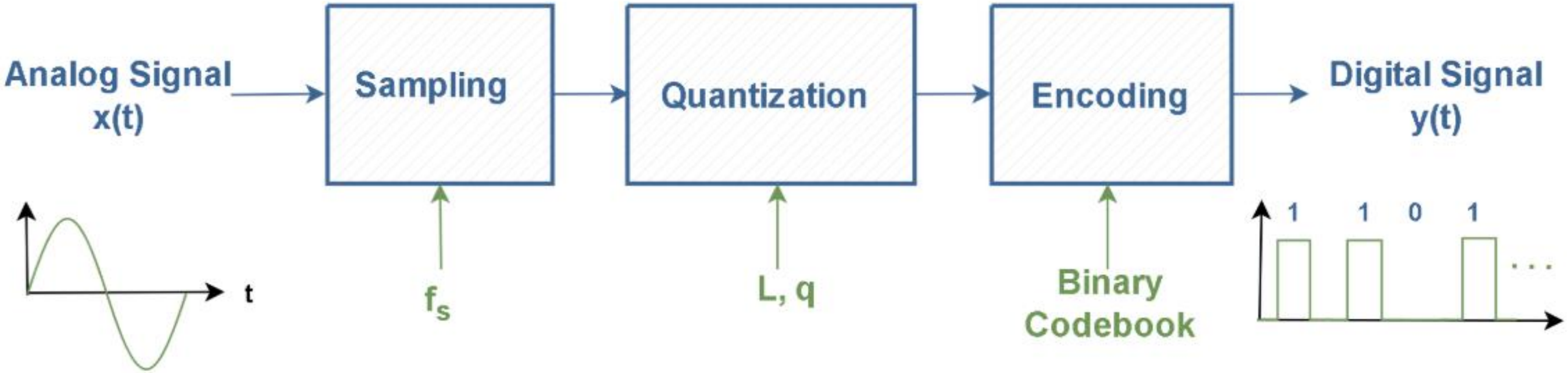
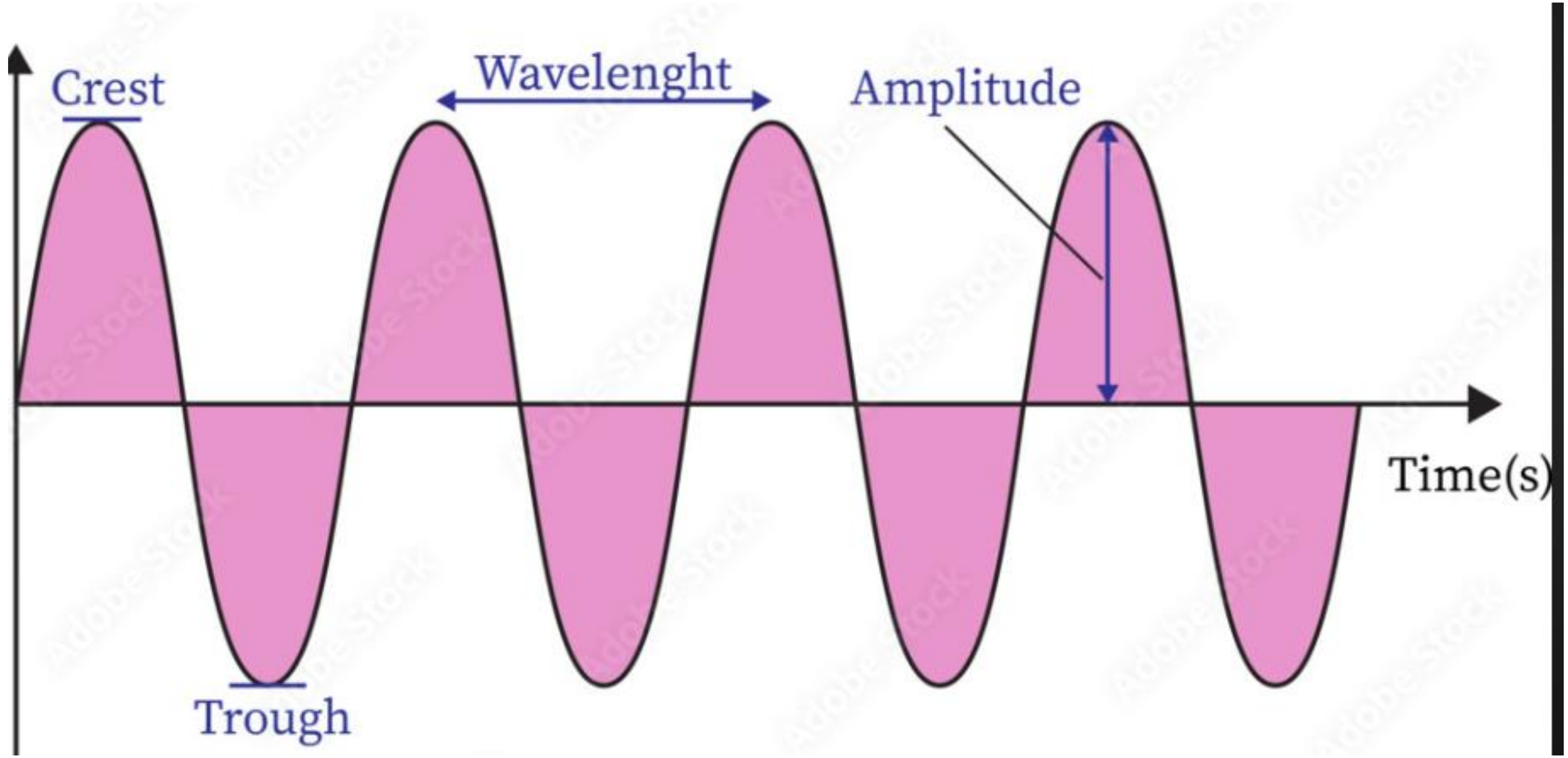
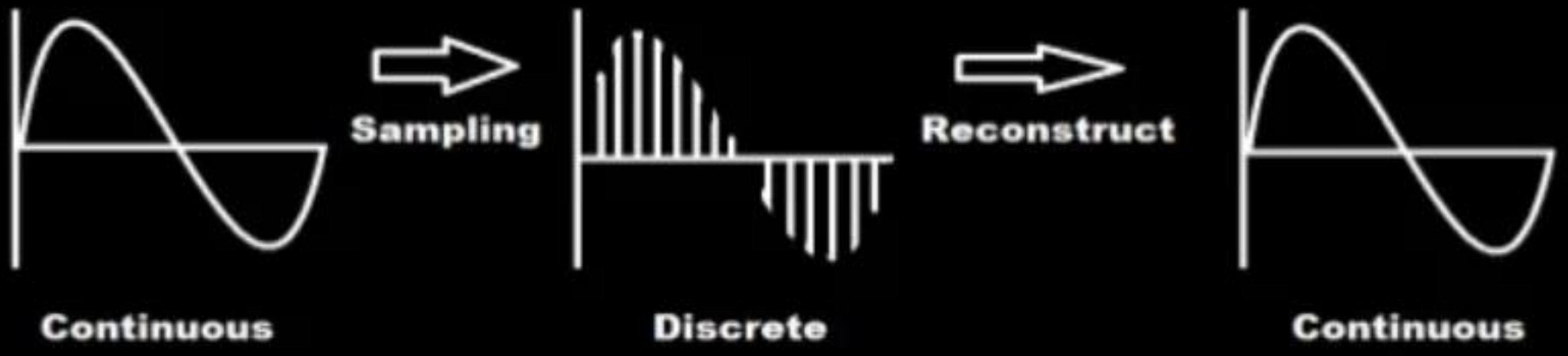


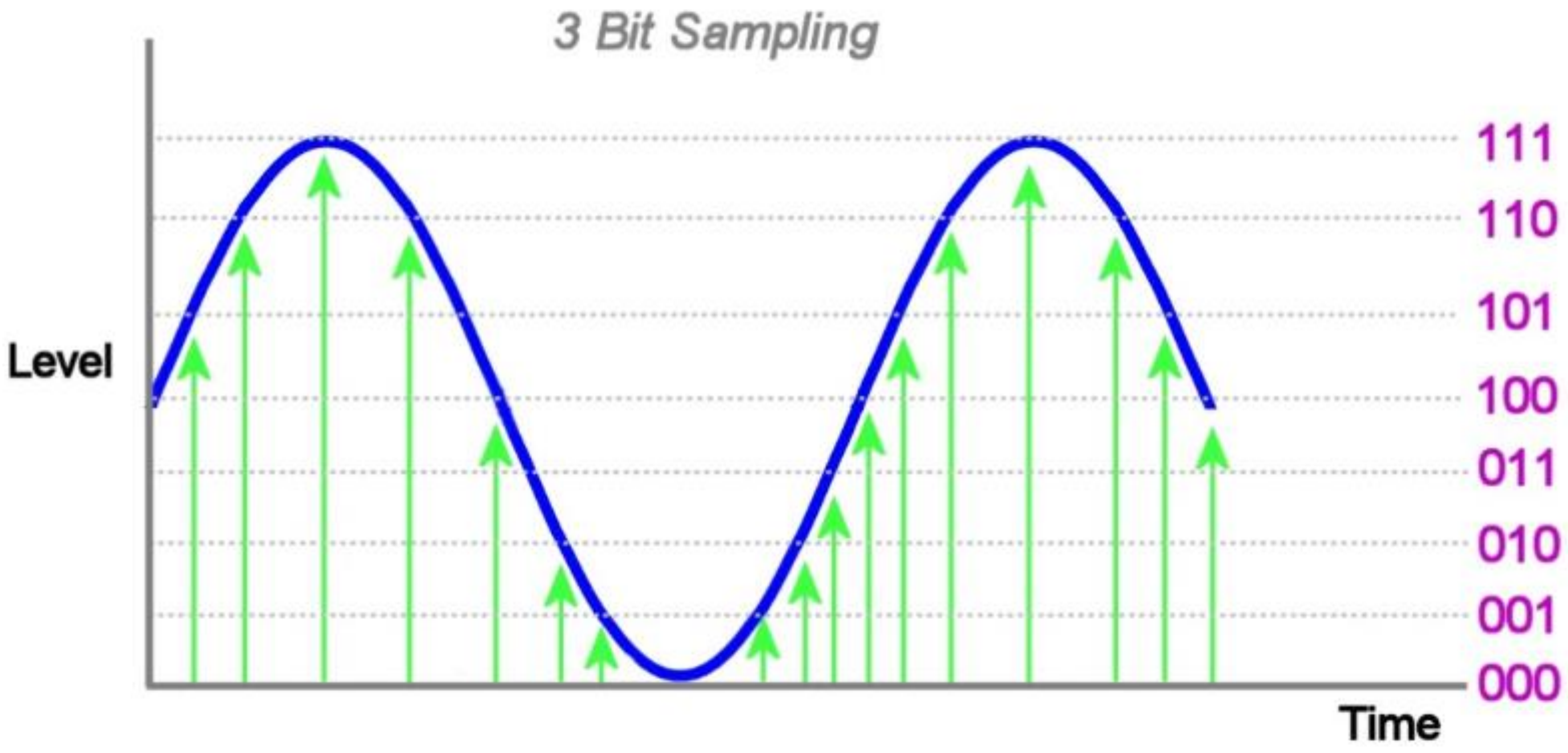
Figure 2: The PCM System

https://www.electronics-lab.com/article/analog-to-digital-conversion-part-1/


Crest
Wavelenght
Amplitude
Time(s)
Trough

- Sampling means splitting in regular interval to convert it in digital form.
- Sampling converts continuous signal to discrete signal.
- Actually the accuracy of analogue signal is more than digital because digital signal have finite data.
- All the positions in discrete signal is a number.



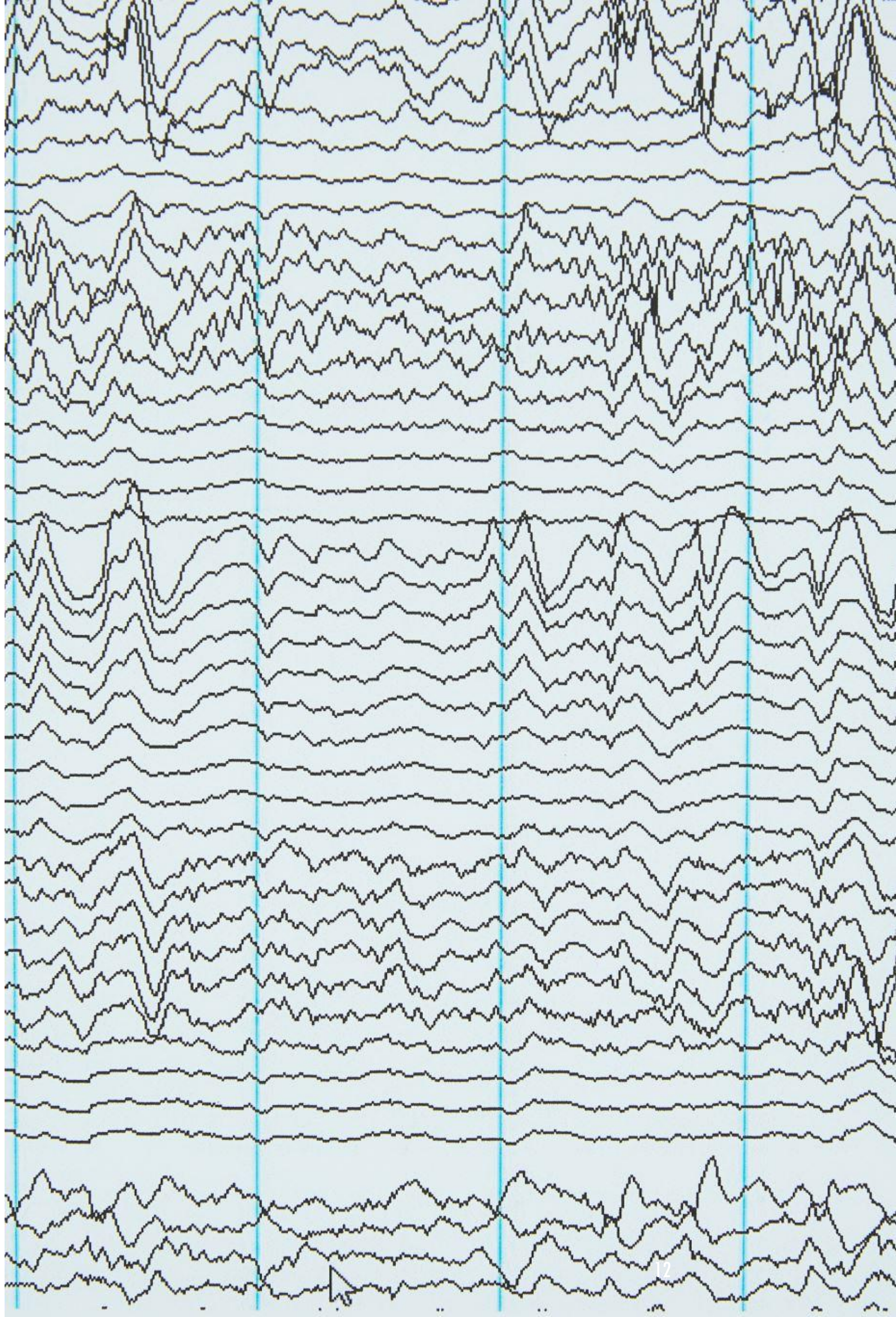


The signal is sampled at regular intervals (shown by the green arrows) and the level stored as a binary number

QUANTIZATION AND AMPLITUDE ERRORS

Quantization involves mapping the amplitude of the sampled audio signal to a finite set of values. This process introduces **quantization errors**, which are the differences between the actual signal amplitude and the quantized value. These errors can manifest as noise in the digital signal.

Amplitude errors occur when the amplitude of the audio signal exceeds the range that can be represented by the quantization levels. This can lead to **clipping**, where the peaks of the audio signal are cut off, resulting in distortion.



SAMPLING AND ERRORS

In practical scenarios, several factors can affect the quality of the sampled audio:

- **Aliasing**: Occurs when the sampling rate is too low to accurately capture the audio signal, causing high-frequency components to be misrepresented as lower frequencies. This can be mitigated by using an appropriate sampling rate and anti-aliasing filters.
- **Jitter**: Variations in the timing of the sampling process can introduce errors in the digital signal. High-quality audio equipment minimizes jitter to ensure accurate sampling.
- **Bit Depth**: Refers to the number of bits used to represent each sample. Higher bit depths (e.g., 16-bit, 24-bit) provide more precise quantization, reducing quantization errors and improving audio quality.

FOURIER TRANSFORM (FT)

Fourier Transform (FT) is used to analyze the frequencies present in a sound wave. Sound waves are composed of a combination of various frequencies, and the FT can separate these frequencies into their individual components.

When a sound wave is recorded, it is first sampled at a specific rate, usually referred to as the sampling rate. The samples are then converted into a digital signal, which can be analyzed using FT. The FT can be used to convert the digital signal from the time domain to the frequency domain, which allows us to see which frequencies are present in the signal and how strong they are.

PHONETICS AND PHONOLOGY

- PHONETICS

- PHONOLOGY AND LINGUISTICS

- SUPRASEGMENTAL FEATURES OF SPEECH

PHONETICS

Phonetics is the study of the physical sounds of human speech.

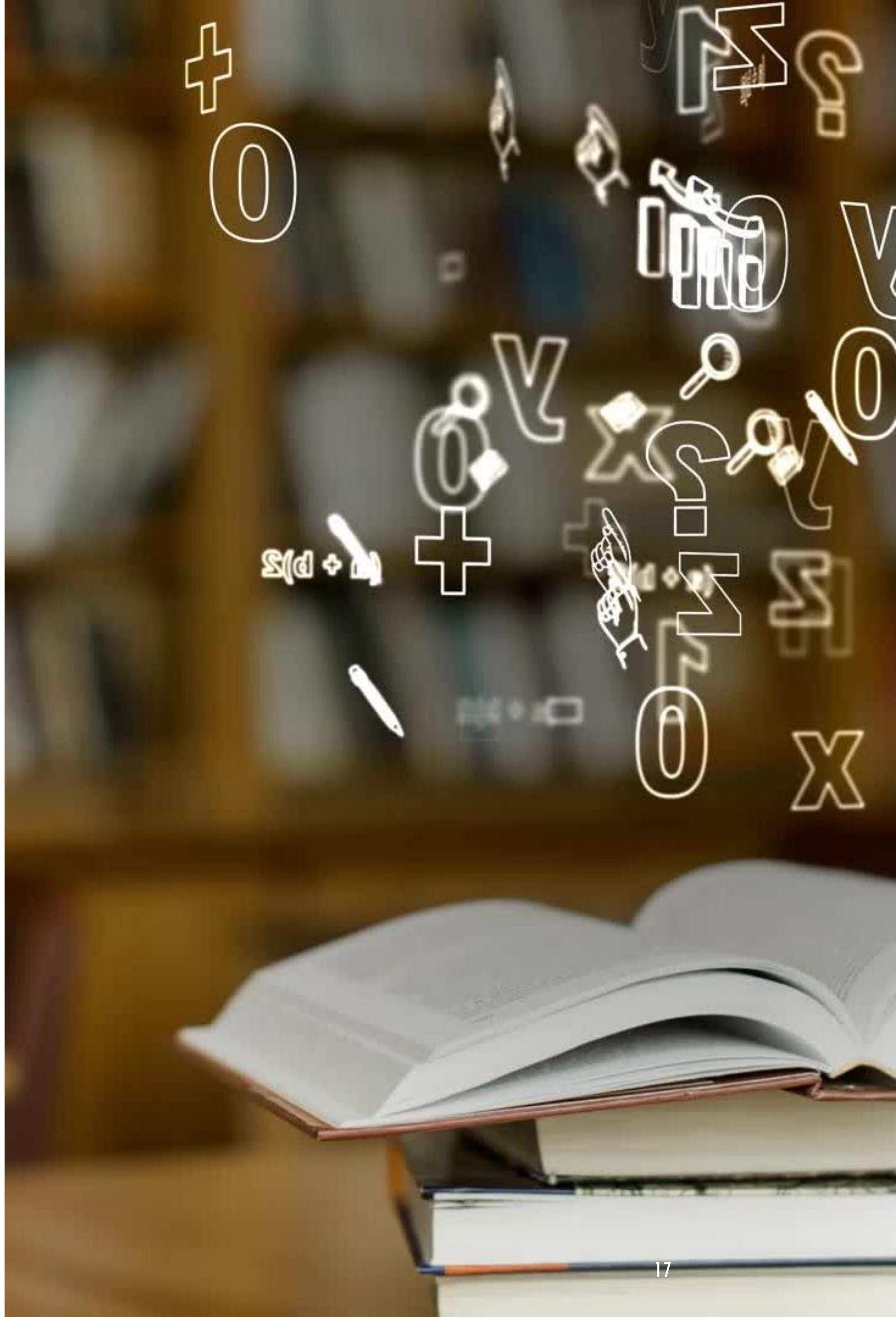
It focuses on how sounds are produced (articulatory phonetics), how they are transmitted (acoustic phonetics), and how they are perceived (auditory phonetics).

Phonetics examines the properties of speech sounds, such as their frequency, duration, and intensity, and how these sounds are articulated by the vocal organs.

PHONOLOGY AND LINGUISTICS

Phonology is the study of how sounds function within a particular language or languages. It deals with the abstract, cognitive aspects of sounds, such as how they are organized in the mind and how they interact with each other.

Linguistics is the broader scientific study of language, encompassing various subfields, including phonetics, phonology, syntax, semantics, and pragmatics. It aims to understand the structure, function, and development of language.



SUPRASEGMENTAL FEATURES OF SPEECH

Suprasegmental features are aspects of speech that go beyond individual sounds (segments) and affect larger units such as syllables, words, or phrases. These features include:

- **Intonation**: The variation in pitch while speaking, which can convey different meanings or emotions (e.g., rising intonation in questions).
- **Stress**: The emphasis placed on certain syllables or words, which can change the meaning of a sentence (e.g., "record" as a noun vs. "record" as a verb).
- **Rhythm**: The pattern of sounds and silences in speech, influenced by the timing and duration of syllables and pauses.
- **Tone**: In tonal languages, the pitch contour of a syllable can change the meaning of a word (e.g., Mandarin Chinese).

LANGUAGE MODELING

- Hidden Markov Modeling

- Neural Networks (TDNN, LSTM, RNN, CNN)

- Other recent machine learning techniques

HIDDEN MARKOV MODELING

Hidden Markov Models (HMMs) are statistical models used to represent systems that have hidden states. These models are particularly useful for sequential data, such as speech recognition. An HMM consists of:

- **States**: Represent the hidden conditions or stages of the system.
- **Observations**: Visible outputs that depend on the hidden states.
- **Transition Probabilities**: Probabilities of moving from one state to another.
- **Emission Probabilities**: Probabilities of observing a particular output given a state.

HMMs are widely used in speech recognition, handwriting recognition, and bioinformatics

NEURAL NETWORKS

Neural Networks are a class of machine learning models inspired by the human brain. They consist of interconnected nodes (neurons) that process information. Key types include:

- **Time-Delay Neural Networks (TDNN)**: Designed to handle temporal sequences by incorporating time delays into the network architecture. TDNNs are effective for tasks like speech recognition where timing is crucial.
- **Long Short-Term Memory (LSTM)**: A type of Recurrent Neural Network (RNN) that can learn long-term dependencies. LSTMs are particularly useful for sequential data, such as language modeling and speech recognition.
- **Recurrent Neural Networks (RNN)**: Designed to handle sequential data by maintaining a hidden state that captures information from previous time steps. RNNs are used in tasks like language modeling and time series prediction.
- **Convolutional Neural Networks (CNN)**: Primarily used for image processing, CNNs can also be applied to speech recognition by extracting features from spectrograms. They use convolutional layers to detect patterns in data.

OTHER RECENT MACHINE LEARNING TECHNIQUES

Recent advancements in machine learning have introduced several powerful techniques:

- **Transformers**: These models use self-attention mechanisms to process sequential data more efficiently than RNNs. Transformers are the backbone of models like GPT-4 and BERT.
- **Generative Adversarial Networks (GANs)**: Consist of two networks (generator and discriminator) that compete against each other to generate realistic data. GANs are used in image synthesis and data augmentation.
- **Reinforcement Learning**: Involves training agents to make decisions by rewarding them for desirable actions. This technique is used in robotics, game playing, and autonomous systems.

APPLICATIONS OF SPEECH RECOGNITION

- Large Vocabulary Speech Recognition
- Keyword recognition
- Speaker Recognition
- Emotion Detection
- Sequence-to-sequence modeling

BASIC SIGNAL PROCESSING

Signal processing is a crucial step in speech recognition, involving several key techniques to convert raw audio signals into a format that can be analyzed and understood by machines. Here are the fundamental concepts:

1. PRE-PROCESSING

Before any analysis, the raw audio signal is pre-processed to enhance its quality and remove noise. This includes:

- **Noise Reduction**: start to reduce background noise.
- **Echo Cancellation**: Removing echoes that can distort the signal.

2. SAMPLING

Sampling is the process of converting a continuous audio signal into a discrete digital signal by taking measurements at regular intervals. The key aspects include:

- **Sampling Rate**: The number of samples taken per second, measured in Hertz (Hz). Common rates are 44.1 kHz and 48 kHz.
- **Nyquist Theorem**: To avoid aliasing, the sampling rate should be at least twice the highest frequency present in the signal.

3. QUANTIZATION

Quantization involves mapping the amplitude of the sampled audio signal to a finite set of values. This process introduces quantization errors, which are the differences between the actual signal amplitude and the quantized value. These errors can manifest as noise in the digital signal.

4. FEATURE EXTRACTION

Feature extraction transforms the raw audio signal into a set of features that can be used for recognition. Common techniques include:

- **Fourier Transform**: Converts the time-domain signal into its frequency-domain representation.
- **Mel-Frequency Cepstral Coefficients (MFCCs)**: Capture the power spectrum of the audio signal, mimicking the human ear's perception of sound.

5. SPECTRAL ANALYSIS

Spectral analysis involves examining the frequency components of a signal. Key methods include:

- **Spectrograms**: Visual representations of the spectrum of frequencies in a signal as it varies with time.

6. SIGNAL ENHANCEMENT

Signal enhancement techniques improve the quality of the speech signal by reducing noise and other distortions. This includes:

- **Filtering**: Removing unwanted frequencies from the signal.
- **Dynamic Range Compression**: Adjusting the amplitude of the signal to maintain a consistent volume level.

7. POST-PROCESSING

After feature extraction, the features are often normalized and transformed to improve recognition accuracy. This can include:

- **Feature Scaling**: Adjusting the range of features to a standard scale.
- **Dimensionality Reduction**: Techniques like Principal Component Analysis (PCA) to reduce the number of features while retaining important information.

These steps form the foundation of signal processing in speech recognition, enabling machines to accurately interpret and respond to human speech.

ACOUSTIC MODELING

Acoustic modeling is a fundamental component of speech recognition systems. It involves creating statistical representations that describe the relationship between phonetic units (like phonemes) and their corresponding acoustic signals. The goal is to accurately capture the variations in speech due to different accents, pronunciations, speeds, and environmental noise. Key aspects include:

- **Feature Extraction**: Transforming raw audio signals into a format suitable for machine learning algorithms. This process compresses essential characteristics of speech while eliminating noise and redundancy.
- **Model Training**: Using large datasets of recorded speech to train models. These datasets often include diverse speakers to ensure robustness and adaptability.
- **Types of Models**: Traditional models like Hidden Markov Models (HMMs) and more recent neural network-based models, such as deep learning architectures, are used. Neural models often yield improved performance due to their ability to learn from vast datasets.

LABELLING

Labelling in the context of speech recognition refers to the process of annotating audio data with corresponding text or phonetic transcriptions. This is crucial for training and evaluating speech recognition models. Key steps include:

- **Manual Labelling**: Human annotators listen to audio recordings and transcribe the speech. This can be time-consuming but ensures high accuracy.
- **Automatic Labelling**: Using pre-trained models to generate initial transcriptions, which can then be refined by human annotators. This speeds up the process and reduces the workload.

Acoustic modeling and labelling are essential for developing accurate and robust speech recognition systems, enabling machines to understand and process human speech effectively.

COMMON ALGORITHMS FOR LANGUAGE MODELING

Language modeling is a crucial aspect of natural language processing (NLP) and involves predicting the next word in a sequence given the previous words. Here are some of the most common algorithms used for language modeling:

1. N-GRAMS

N-grams are a simple and traditional method for language modeling. An N-gram model predicts the next word in a sequence based on the previous $N-1$ words. For example, a bigram model (2-gram) uses the previous word to predict the next word, while a trigram model (3-gram) uses the previous two words.

N-gram models are easy to implement but can struggle with long-range dependencies due to their limited context.

2. HIDDEN MARKOV MODELS (HMMS)

Hidden Markov Models are statistical models that represent the probabilities of sequences of observed events, such as words in a sentence.

HMMs are particularly useful for tasks like part-of-speech tagging and speech recognition. They model the sequence of words as a series of hidden states, each emitting an observable word

3. RECURRENT NEURAL NETWORKS (RNNS)

Recurrent Neural Networks are a type of neural network designed to handle sequential data. RNNs maintain a hidden state that captures information from previous time steps, making them suitable for language modeling.

However, traditional RNNs can suffer from vanishing gradient problems, which limit their ability to learn long-range dependencies.

4. LONG SHORT-TERM MEMORY NETWORKS

LSTM networks are a special type of RNN designed to overcome the limitations of traditional RNNs. LSTMs use gates to control the flow of information, allowing them to maintain long-term dependencies and capture context over longer sequences. This makes them highly effective for language modeling.

5. GATED RECURRENT UNITS (GRUS)

GRUs are another variant of RNNs that simplify the architecture of LSTMs while retaining their ability to capture long-term dependencies. GRUs use fewer gates than LSTMs, making them computationally more efficient while still performing well on language modeling tasks.

6. TRANSFORMERS

Transformers are a more recent and powerful architecture for language modeling. They use self-attention mechanisms to process entire sequences of words simultaneously, rather than sequentially. This allows transformers to capture long-range dependencies more effectively and efficiently. Transformers are the backbone of state-of-the-art models like GPT-4 and BERT.

7. BIDIRECTIONAL ENCODER REPRESENTATIONS FROM TRANSFORMERS

BERT is a transformer-based model that pre-trains deep bidirectional representations by conditioning on both left and right context in all layers. This allows BERT to achieve high performance on a variety of NLP tasks, including language modeling.

8. GENERATIVE PRE-TRAINED TRANSFORMERS

GPT models, such as GPT-3 and GPT-4, are transformer-based models pre-trained on large corpora of text.

They generate coherent and contextually relevant text by predicting the next word in a sequence. GPT models are highly effective for language modeling and text generation.

DECODING ACOUSTIC FEATURES INTO SPEECH

Decoding acoustic features into speech involves converting the processed audio signals into text or commands that a machine can understand. This process is a critical part of speech recognition systems and includes several key steps:

1. Feature Extraction

Before decoding, the raw audio signal is transformed into a set of acoustic features. Common techniques include:

- **Mel-Frequency Cepstral Coefficients (MFCCs)**: Capture the power spectrum of the audio signal, mimicking the human ear's perception of sound.
- **Linear Predictive Coding (LPC)**: Models the vocal tract to represent the spectral envelope of the speech signal.
- **Spectrograms**: Visual representations of the spectrum of frequencies in a signal as it varies with time.

2. Acoustic Modeling

Acoustic models establish the relationship between the extracted features and phonetic units (like phonemes). These models are trained using large datasets of recorded speech to ensure robustness and adaptability. Common models include:

- **Hidden Markov Models (HMMs)**: Statistical models that represent sequences of observed events, such as words in a sentence.
- **Neural Networks**: Deep learning architectures, such as Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which can learn complex patterns in speech data.

3. Decoding Algorithms

Decoding algorithms convert the output of the acoustic model into a final transcription. Common algorithms include:

- **Greedy Search**: A simple decoding algorithm that selects the most likely output at each step.
- **Beam Search**: A more sophisticated algorithm that considers multiple possible outputs at each step and selects the most likely sequence.

4. Language Modeling

Language models predict the likelihood of sequences of words, helping to improve the accuracy of the decoded speech. Common models include:

- **N-grams**: Predict the next word in a sequence based on the previous N−1$N-1$ words.
- **Neural Networks**: Models like Long Short-Term Memory (LSTM) networks and Transformers, which can capture long-range dependencies and context.

5. Post-Processing

After decoding, the transcriptions may be further refined to improve accuracy. This can include:

- **Error Correction**: Using context and grammar rules to correct mistakes in the transcription.
- **Normalization**: Adjusting the format of the text to match expected outputs (e.g., converting numbers to words).

PYTHON PACKAGES

Python packages for speech recognition:

- **SpeechRecognition**
- **PyDub**
- **PyAudio**
- **Vosk**
- **OpenAI Whisper**
- **DeepSpeech**
- **Transformers**
- **Google Cloud Speech-to-Text**
- **Azure Cognitive Services Speech SDK**
- **Wit.ai**

CASE STUDIES

https://medium.com/@afkhoso/case-study-speech-recognition-fbc420e69962

https://github.com/Rpersie/Speech-Recognition

RESOURCES

https://medium.com/